

Tool-Augmented Force-Position Control for a Quadrupedal Mobile Manipulator Using Reinforcement Learning

Interim Report and Project Plan - Thesis A

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Abstract

Quadrupedal mobile manipulators offer a way of carrying out manipulation tasks in areas that may be difficult to access with fixed robotic arms or wheeled platforms. Contact-based tasks are however still challenging, as the robot must coordinate its balance, movement, arm position and applied force at the same time. This thesis will investigate whether a force-aware whole-body reinforcement learning policy can allow a Unitree Go2 EDU with a D1 arm to complete sustained-contact tasks using different external tools. The policy is intended to receive desired tool-tip position and force commands, while a higher-level task interface will provide information such as the required trajectory, selected tool and task objective. Development will involve comparing force-aware control against position-only control, as well as testing the use of tools against the bare end effector. Tool-conditioned and unconditioned policies will also be compared to investigate whether providing information about the attached tool improves performance across different tasks. Training and initial testing will be carried out in Isaac Sim and Isaac Lab before the policy is progressively transferred to the physical robot. The experiments will assess whether force awareness and tool conditioning improve contact stability, task completion and the ability to operate with more than one tool. At the time of this report, the combined Go2 and D1 simulation model has been established, together with the required ROS 2 interfaces and preliminary locomotion testing. This provides the starting point for developing and evaluating the baseline policies.

Nomenclature

Term / Symbol	Definition
F_{cmd}	Commanded interaction force
F_{ext}	External/contact force
F_{react}	Estimated or measured reaction force
x_{cmd}	Commanded end-effector position
x_{des}	Desired end-effector or tool-tip position
x_{target}	Modified position target supplied to the controller
K, D, M	Stiffness, damping and equivalent inertia

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1 Introduction

1.1 Motivation

Quadrupedal mobile manipulators combine the movement capability of a legged robot with the reach and flexibility of a robotic arm. The mobile base allows the complete system to move around obstacles, adjust the position and orientation of the arm, and reach areas that would not be accessible from a fixed stance. Previous whole-body learning approaches have shown that the legs, body and arm can be coordinated together for reaching, grasping and object interaction, rather than treating each part as a separate system [5], [8], [15].

This however becomes more difficult when the end effector must remain in contact with the environment. A position-only controller may follow a trajectory accurately in free space, but changes in the surface position, stiffness or friction could cause contact loss or robot base imbalance. Force-aware and hybrid force-position approaches attempt to account for this by including interaction force as a command or estimated value, rather than allowing it to result only from position error [24], [38].

External tools provide a practical way of expanding the tasks that the robot can complete without replacing the complete arm or end effector. For example, a probe can apply force over a smaller contact area, while a wiping pad can maintain contact across a wider surface. A scraper may need to resist loads along the surface, and a lever or wrench can convert the movement of the end effector into rotation. The use of these tools however also adds further control challenges. The actual point of interaction is moved away from the gripper, and the mass, shape, inertia and stiffness of the attachment will affect the forces passed through the arm and into the robot body. It follows that tool design cannot be considered completely separately from the controller [16].

Reinforcement learning is suitable for this type of problem because coordination can be learned across the complete robot, while exposing the policy to different contact conditions and tool properties in simulation. This does not remove the need of careful development however. Reward functions, actuator behaviour, contact modelling, safety restrictions and the process of transferring the policy to the physical system will all affect the result. Previous work on legged robots has shown in particular that realistic actuator modelling and robust simulation are important for successful transfer to hardware [10]. These considerations support beginning with simpler baseline policies before progressing towards force-aware and tool-conditioned whole-body control on the Go2-D1 platform.

The main technical problem is coordinating the legs, floating base and D1 arm while the functional tool tip follows a required path and maintains the desired contact with an uncertain surface or mechanism. At the same time, the robot must remain balanced, stay within its joint and actuator limits, and account for the forces and moments passed through the attached tool. This becomes more difficult when the geometry and physical properties of the tool change between tasks. The important interaction frame is also located at the functional tip of the tool rather than at the gripper itself. As a result, an incorrect tool registration or payload model could produce both position and force errors, even where the motion of the gripper appears to be accurate.

Therefore, the research problem is not only whether a quadrupedal manipulator can complete a tool-use demonstration, but whether force awareness and explicit tool information provide a quantifiable improvement over simpler baselines. The selected metrics should separate basic task completion from aspects such as force tracking, continued contact and whole-body safety. The resulting policy must also remain sufficiently robust for a staged transfer from Isaac Sim to the physical Go2 EDU and D1 arm.

1.2 Research Objectives

Research presented will aim to critically analyse methods of whole body control, force-awareness and architectures of control policies with the intent of satisfying the following research objectives:

- Develop a whole-body reinforcement learning controller that accepts position and force-related commands for contact-rich manipulation.
- Integrate specified external tools into the robot model, observation space and task formulation.
- Compare force-aware control with an otherwise equivalent position-only baseline.
- Compare task performance with and without the external tool.
- Compare a tool-conditioned policy with a non-tool-conditioned policy.
- Evaluate simulation performance and transfer the selected policy to the physical robot.

Autonomous tool search, semantic tool selection and acquisition of freely positioned tools are outside the core scope so that the control problem can be evaluated without conflating it with perception and grasp-planning failures.

2 Literature Review

2.1 Quadrupedal Mobile Manipulation and Loco-Manipulation

Mobile manipulation combines mobile platforms with manipulators such as robotic arms to extend object interaction beyond a fixed workspace. Legged robots in particular illustrate that this mobility and manipulation are tightly coupled, as an optimal response to a task may involve repositioning the feet, body and end effector. This coupling is referred to as loco-manipulation, and it relates to how the dexterity and traversal benefits offered by a legged robot introduce a substantially more difficult control problem: end-effector motion must be coordinated with balance, locomotion, foothold selection and the reaction forces transmitted through the arm and robot body [38].

Of particular difficulty is how, unlike a fixed-base manipulator, the underactuated floating base of a quadrupedal mobile manipulator cannot be controlled directly. Its motion is instead governed by contact forces through its feet and end effector, a particular challenge for controllers because arm acceleration, body posture, and forces applied to the environment often alter the support forces required to maintain balance. Furthermore, walking and manipulation produce changing combinations of active contacts, meaning that loco-manipulation must account for both continuous motion and transitions between different contact conditions.

Early quadruped loco-manipulation systems tackled this coupling through explicit model-based planning and whole-body control. An early example using a quadruped and 6-DOF arm is ALMA, which implemented motion planning alongside hierarchical whole-body optimisation to execute arm tasks while walking with dynamic gaits [2]. Importantly, this demonstrated that manipulation could be performed without requiring the quadruped to first stop in a fixed posture. Subsequently, works such as Sleiman et al. formulated locomotion and manipulation as a single multi-contact optimal-control problem [31]. Their model-predictive control framework augmented the robot's centroidal dynamics with the dynamics of the manipulated object, allowing locomotion, end-effector tracking and interactions such as pushing or pulling a resistant door to be represented within the same optimisation problem.

More complex loco-manipulation tasks require not only continuous trajectory optimisation but also decisions about when and where contact should occur. Sleiman et al. addressed this problem by

combining task-and-motion planning, graph search and trajectory optimisation to jointly determine whole-body trajectories and contact schedules [32]. The resulting framework produced both prehensile and non-prehensile behaviours for tasks such as operating heavy dishwashers and traversing spring-loaded doors. This work illustrates that useful loco-manipulation may involve several distinct contact phases: approaching an object, establishing contact, applying or resisting force, repositioning the body and terminating contact. These planning-based approaches, however, demand accurate environmental geometry and task models, which can limit their direct use in uncertain or visually changing environments.

Learning-based approaches provide an alternative in which coordinated behaviour is acquired through interaction rather than being fully specified through analytical models and manually designed state machines. Deep Whole-Body Control demonstrated that learned coordination could enable a quadruped to alter its body posture and locomotion in support of an end-effector task, producing behaviours such as leaning, crouching and repositioning the base to extend the arm’s effective workspace [5]. Visual Whole-Body Control similarly demonstrated the benefit of posture as a component of the manipulation workspace by noting the improved ability to reach objects across different heights and positions [15]. These results highlight that the practical workspace of a legged manipulator is determined by the coordinated motion of the complete robot rather than by the kinematic reach of the arm alone.

Whole-body coordination is also important before contact occurs. QuadWBG’s introduction of a Generalised Oriented Reachability Map enabled the assessment of candidate 6-DOF base poses relative to a desired grasp pose, allowing the robot to position its body in configurations that improved arm reachability and supported grasping across a large vertical workspace [35]. A reported 89% one-attempt grasping success rate in its real-world experiments shows that loco-manipulation is not simply locomotion followed by manipulation; the locomotion objective must often anticipate the subsequent grasp or interaction. For tool-mediated tasks, the selected base configuration may need to support both tool acquisition and later use. A posture suitable for grasping a tool may not provide sufficient stability, reach or force transmission for pressing, scraping or driving the tool against a workpiece.

Quadrupedal manipulation does not always require a dedicated arm. Pedipulate demonstrated that one of the robot’s legs could instead be used as an end effector while the remaining legs maintained balance [1]. Its learned controller produced whole-body behaviours for door opening, sample collection and object pushing, including locomotion when targets lay beyond the stationary workspace. Although this configuration differs from the arm-equipped platform considered in this thesis, it reinforces the broader principle that manipulation capability arises from coordinated use of the entire body rather than from an isolated manipulator.

Overall, the literature demonstrates that quadrupedal loco-manipulation extends manipulation capability through active body repositioning, terrain traversal and the use of multiple contacts. Model-based approaches analyse dynamics, constraints and contact schedules, while learned approaches have demonstrated adaptable behaviours without requiring every interaction to be described analytically. However, the reviewed systems predominantly address reaching, grasping, object transport or direct environmental interaction. For tool-mediated manipulation, loco-manipulation must additionally account for the altered reach, contact geometry, payload, inertia and force direction introduced by the selected tool. Achieving these capabilities requires an appropriate whole-body control architecture, which is examined in the following section.

2.2 Whole-Body Control for Legged Manipulators

Whole-body control refers to the coordinated use of a robot's available degrees of freedom to achieve task objectives while maintaining dynamic stability. For a quadrupedal manipulator, this requires simultaneous consideration of the legs, floating base and robotic arm rather than treating the manipulator as an independent system mounted on a stationary platform. Arm movement changes the robot's centre of mass and produces reaction forces and moments that must be resisted through the feet. Conversely, changes in body posture and foot placement alter the manipulator's reachable workspace, joint configuration and ability to apply force. Whole-body control must therefore balance manipulation performance against locomotion stability, contact constraints, joint limits and actuator capabilities.

Classical whole-body controllers generally formulate this coordination as a constrained optimisation problem. ALMA used a hierarchical model-based architecture in which motion planning generated desired trajectories and a whole-body controller distributed the required motion across the quadruped and manipulator [2]. Sleiman et al. subsequently represented locomotion, manipulation and object interaction within a unified model-predictive control formulation [31]. This allowed end-effector objectives, base motion, contact forces and dynamic constraints to be considered within the same optimisation problem rather than being resolved by separate controllers. The primary advantage of these approaches is that physical constraints, including friction cones, contact conditions and torque limits, can be explicitly enforced. However, their effectiveness depends on the accuracy of the robot, actuator, object and contact models.

Model-based whole-body controllers may also regulate the robot's mechanical response during contact. Risiglione et al. incorporated Cartesian impedance objectives for both the quadruped base and manipulator within a quadratic-programming whole-body controller [27]. This demonstrated that compliance is not solely an arm-level property because forces transmitted through the end effector influence the complete robot and must be reconciled with the support forces applied at the feet. However, analytical controllers commonly rely on simplified dynamic or impedance models to enable real-time optimisation. These simplifications may become less representative when an external tool changes the effective end-effector geometry, payload and contact dynamics.

As a concept, learning-based whole-body control rejects the analytically accurate models and manually tuned coordination rules of Model based approaches. Deep Whole-Body Control [5] identifies the limitation of these conventional modular systems as the difficult coordination of independently developed arm and locomotion controllers, stating that errors or incompatible objectives may propagate between modules and produce non-smooth or unnatural motion. The authors instead trained a single reinforcement-learning policy over the complete quadruped-manipulator action space. By learning across the full body, the robot could alter its posture and locomotion to support the end-effector task rather than assuming that the base would remain fixed during manipulation.

Although this unified formulation reduces the need to define interfaces between separate controllers, its high-dimensional action space creates a more difficult optimisation problem. Deep Whole-Body Control therefore introduced specialised learning and adaptation techniques to support policy training and deployment [5]. However, its objectives remained predominantly position and locomotion-based, meaning that environmental force was produced as a consequence of task execution rather than being treated as an independently controllable input.

Zhi et al. extended the unified-policy concept by incorporating position and force commands within the same low-level whole-body controller [38]. This retained the benefit of a single execution policy while expanding its command space to include position tracking, force application and compliant interaction. The approach is particularly relevant to tool-mediated tasks because the controller may need to

coordinate whole-body posture while regulating motion along one direction and interaction force along another. The detailed distinctions between position, force, hybrid force-position and impedance control are discussed in Section 2.3.

A unified whole-body controller does not require perception, planning and motor control to be learned within one monolithic network. Visual Whole-Body Control used a hierarchical architecture in which a visual policy generated manipulation goals while a reinforcement-learned whole-body controller coordinated the legs, body and arm [15]. This separated comparatively slow visual task reasoning from high-frequency balance and motor coordination. Its jointly trained non-hierarchical baseline failed to learn the evaluated grasping tasks, whereas the hierarchical system achieved success across several object categories, indicating that task decomposition can make learning more tractable.

UMI-on-Legs adopted a similar separation between task-level manipulation and embodiment-specific execution [8]. Its manipulation policy generated an end-effector trajectory in a task-centred frame, which was tracked by a quadruped-specific whole-body controller. This enabled manipulation policies developed independently of the complete quadruped to be deployed through a specialised execution layer. These systems demonstrate that hierarchical architecture is compatible with unified whole-body control: task interpretation can remain modular while the high-frequency controller coordinates the complete robot.

Hierarchical systems nevertheless introduce interfaces through which errors may propagate. Perception error, trajectory-prediction error, coordinate-frame misalignment and low-level tracking error may accumulate across modules. Furthermore, an end-effector trajectory alone does not specify how strongly the robot should interact with the environment. This is particularly limiting for tool-mediated operations, in which the same tool-tip trajectory may fail if the normal force is too small or damage the workpiece if excessive force is applied. A suitable high-level interface may therefore need to include both a tool-tip pose or trajectory and a desired force or compliance profile.

Hybrid architectures combine learned and model-based components to retain some advantages of each approach. Schperberg et al. combined model-based manipulation control with reinforcement-learned locomotion on a quadrupedal manipulator [29]. The analytical component provided interpretable compliance and safety constraints, while the learned locomotion component coordinated the legs and base under uncertain operating conditions. Such architectures can retain explicit physical constraints while using learning for high-dimensional behaviours that are difficult to model or engineer directly. However, they also introduce additional interfaces and dependencies between the analytical and learned subsystems.

The main advantages and limitations of these architectures are summarised in Table 1.

Table 1: Summary of whole-body control architectures for quadrupedal manipulators.

Architecture	Main strength	Main limitation
Decoupled	Subsystems can be developed and tested independently.	Requires carefully engineered interfaces and may produce conflicting objectives [5].
Model-based unified	Explicitly represents dynamics, contacts and physical constraints.	Depends on accurate models and real-time optimisation [2], [27], [31].
Unified reinforcement learning	Learns coordinated full-body behaviour with fewer manually designed rules.	High-dimensional training and safety validation remain challenging [5], [38].
Hierarchical or hybrid	Separates task reasoning, force regulation and fast whole-body execution.	Introduces interface complexity and potential error propagation [8], [15], [29].

For this thesis, these findings support a hierarchical architecture with a unified low-level force-position controller. A high-level task or tool policy can specify the required tool-tip pose, trajectory and interaction-force objectives, while the whole-body controller coordinates the legs, base and arm to maintain balance, extend reach and transmit the commanded force through the selected tool. The required position, force and compliance commands are examined in the following section.

2.3 Position, Force, Hybrid Force-Position and Impedance Control

Position control is the regulation of end effector location or trajectory without the specification of any contact forces. It is often applied to free-space reaching and grasping, however in contact-rich scenarios such as wiping, scraping or pressing, physical tool and surface errors are usually present alongside controller miscalibration. In these scenarios where success depends on movement and maintained force, loss of contact or unexpected large forces are common.

Classical hybrid position-force control handles this limitation by separating task-space motion into constrained and unconstrained directions [26]. It entails the control of force along axes relevant to environmental contact, and allows position-only control along other accepted directions. For example, a wiping task necessitates force control normal to the surface, but trajectory control tangentially. An important extension to this formulation was later made by Yoshikawa, accounting for manipulator dynamics along with end-effector constraints [37]. This is of particular relevance to defining the inertia of tools.

Impedance control instead regulates the relationship between motion and interaction force rather than independently commanding exact position and force. Hogan represented this behaviour through a desired mass-spring-damper relationship [9]:

$$F = K(x - x_{des}) + D(\dot{x} - \dot{x}_{des}) + M(\ddot{x} - \ddot{x}_{des})$$

where K , D and M represent desired stiffness, damping and equivalent inertia. A low stiffness allows the end effector to yield more readily to external disturbance, while a high stiffness produces stronger resistance to displacement. Impedance control is therefore useful where contact geometry is uncertain, as it permits the robot to remain compliant while still tending towards a commanded trajectory. At the whole-body level, Risiglione et al. demonstrated that Cartesian impedance objectives can be imposed on both the quadruped base and manipulator while satisfying friction, contact, joint and torque constraints [27]. Their results also show that the realised response changes with the robot's foot-contact condition, meaning that compliance during stance cannot be assumed to remain identical while walking.

Admittance control uses the inverse relationship: measured external force is converted into a commanded motion. Schperberg et al., for example, mapped a measured external wrench into desired end-effector velocity and coordinated the response through the arm and legs [29]. This approach is suitable when accurate force measurements are available and allows explicit safety limits to be added. Its reliance upon force sensing differs from learned force-control methods that estimate interaction forces from proprioceptive history.

Portela et al. demonstrated learned direct force control on a legged manipulator without requiring force measurements during deployment [24]. Their policy includes a mode input that switches between position and force objectives during training, which is an interesting technique but importantly sacrifices the ability to have both motion and force as part of the same command - a trouble for the complex tool-oriented tasks on the horizon of this thesis. However, the policy does show an innovation in zero-force tracking, allowing the arm to passively comply with human guidance but hold its position after, which could be of use in collecting demonstrations manually.

The progression towards a unified hybrid formulation can be seen later in Zhi et al., which integrates both position and force in the same policy. Their simplified relationship modifies the end-effector target according to commanded, reaction and external forces:

$$X_{target} = X_{cmd} + [F_{ext} + (F_{cmd} - F_{react})] / K$$

Different combinations of position and force commands can produce tracking, direct force control, force tracking, impedance-like compliance and hybrid force-position behaviour. The flexibility of reusing the same controller for different modes is beneficial to a sequence-based approach to tool operation, where, for example, a position control can guide the approach, normal force can establish contact, tangential position commands can execute the task, and compliant behaviour can be used during disturbance or withdrawal.

The simplification in [38] however omits damping and inertia terms as a simplification for slow interactions, and notes degradation in performance during high-frequency interaction or at workspace boundaries. This is not ideal for handling the impacts, slipping or fast rotational tasks that come with tool manipulation, and consequently the controller developed in this thesis should initially target controlled, sustained-contact tasks rather than high-impact tool use. Another learning from Zhi et al is a warning about motor gain trade-off: lower gains permit more observable displacement and can improve force estimation, but excessively low gains reduce position accuracy.

For tool-mediated manipulation, all motion and force objectives should be defined at the functional tool frame or tool centre point rather than at the gripper alone. Correct gripper motion can still produce an incorrect tool-tip position, orientation or contact force if the grasp transform is not represented. The proposed controller should therefore emulate position, force and compliant modes at the tool tip while coordinating the resulting loads through the complete quadruped-manipulator system.

2.4 Reinforcement Learning for Contact-Rich Loco-Manipulation

Reinforcement learning treats control as an ongoing interaction between a policy and its environment. A policy will, at each timestep, observe the current state, select an action, and receive a reward indicating how well the outcome aligns with the task. In legged manipulation, the observation typically includes the robot’s base orientation and velocity, joint positions and velocities, previous actions, gait phase and task commands [24], [38]. Rather than commanding motor torques directly, many policies output residual joint-position targets that are tracked by lower-level proportional-derivative controllers. This limits the action space and preserves some actuator-level stability while still allowing the policy to coordinate the robot’s entire body.

A major difficulty is that locomotion, manipulation, force regulation and safety operate over different scales and timescales. For example, a policy may improve end-effector accuracy by adopting an unstable posture, reduce force error by driving joints towards saturation, or remain balanced while making little progress on the manipulation task. Deep Whole-Body Control addresses this high-dimensional optimisation problem using Advantage Mixing, which leverages the causal relationship between arm and leg actions, together with Regularised Online Adaptation for deployment [5]. The results demonstrate that a unified policy is possible, but they also highlight that reinforcement learning will not necessarily find a sensible balance between competing objectives without careful design.

Reward design must therefore separate task success from requirements such as safety, efficiency and smooth motion. Zhi et al. reward end-effector position and base-velocity tracking while exposing the robot to sampled force commands and external disturbances. At the same time, they penalise collisions, joint-limit violations, torque, acceleration and rapid changes in action [38]. Portela et al. also include an

explicit force objective rather than expecting contact behaviour to emerge from position tracking alone [24]. This distinction is particularly important for tool use. Reaching the correct tool-tip pose does not guarantee that contact is stable, that the desired normal force has been applied or that the tool is not slipping.

A staged curriculum can make this optimisation problem more manageable. Zhi et al. begin by training whole-body reaching and locomotion before introducing random force commands and external disturbances [38]. A similar approach would suit the proposed system. Introducing tool acquisition and contact before the robot can reliably balance and reach may lead to sparse or conflicting reward signals. Training could instead begin with bare end-effector position tracking, followed by direct contact under commanded forces, manipulation with a pre-attached tool, and finally the generalisation to using multiple tools.

Training can also make use of information that will not be available on the physical robot. In [38], simulator ground truth is used to supervise force and state estimation, whereas the deployed actor relies on task commands and a history of proprioceptive measurements. This asymmetric arrangement allows privileged information to accelerate training without making it a required input during deployment. The same distinction should be maintained in this thesis. Exact contact forces, tool poses and object states may be used during simulation, while the deployed policy should depend only on measurements available from the robot’s joint encoder publishers, IMU, cameras and any force sensor included in the final system.

Most reinforcement-learning systems for legged robots use Proximal Policy Optimisation because its clipped objective reduces the likelihood of large, destructive policy updates between iterations [30]. The method also benefits from massively parallel simulation. Rudin et al., for example, trained thousands of simulated quadrupeds simultaneously on a single GPU and gradually increased terrain difficulty through a curriculum [28]. Isaac Lab provides a similar workflow through GPU-parallel physics, actuator modelling, multi-rate sensors, data-collection tools and domain randomisation [18].

The problem may be simplified further by separating decisions that occur at different timescales. Visual Whole-Body Control uses a slower visual component to generate goals for a high-frequency whole-body controller [15]. Similarly, UMI-on-Legs provides trajectory previews from a slower manipulation policy to a controller trained in simulation [8]. C. Chi et al. add that at this task level, a policy such as Diffusion could be implemented to a great effect to produce multimodal action sequences. It is however, less appropriate for rapid balance correction and force regulation [4]. The literature therefore supports a hierarchical structure for this thesis. A slower tool and task policy would generate tool-tip pose and force objectives, while a faster reinforcement-learned whole-body controller would execute them and maintain stability.

2.5 Force Sensing, Force Estimation and Force-Aware Policies

Force-aware manipulation requires an observation that indicates how the robot is interacting with the environment. A wrist-mounted six-axis force/torque sensor provides direct measurement of the forces and moments transmitted between the manipulator and tool. This information is physically interpretable and useful for calibration, safety limits and evaluation. Schperberg et al. use such a sensor with model-based admittance control and a Reference Governor to produce compliant behaviour with explicit safety constraints [29]. The disadvantages are specifically additional cost, mass, wiring and mechanical complexity at the wrist.

An alternative is to estimate interaction forces using the robot’s own proprioceptive measurements. Portela et al. use joint encoders, IMU data and a history of previous observations to infer external forces,

allowing the policy to follow force commands without requiring a dedicated sensor during deployment [24]. A similar approach is taken by Zhi et al., where a history of base and joint states are used to estimate external force together with end-effector position and base velocity [38]. In their physical experiments, the force-estimation error was approximately 5–10 N over the tested range, with the aforementioned degradation toward workspace borders and in high-frequency interactions.

Observation history is needed because the robot’s instantaneous state alone may not reveal whether a change in motion was caused by external contact, actuator dynamics or a temporary disturbance. Changes in position, velocity and commanded action over time provide additional evidence of contact, although this comes at the cost of delay and greater dependence on the accuracy of the training model. Reducing the motor gains can make contact easier to detect because an applied force produces a larger displacement. However, making the system too compliant also worsens position tracking [38]. Proprioceptive estimation therefore avoids additional sensing hardware, but generally provides force information that is less precise and more dependent on the robot model.

Tactile sensing provides a third approach by observing local deformation, pressure distribution or slip at the gripper and tool interface. Kang et al. separate slower arm-pose prediction from a high-frequency reactive tactile force loop, showing that force regulation may need to operate faster than the visual task policy [11]. TacEx demonstrates that GelSight-style deformation and images can be simulated within Isaac Sim and used in Isaac Lab reinforcement-learning environments [17]. However, it also illustrates that realistic tactile simulation requires soft-contact and sensor-image modelling beyond the rigid-body forces available directly from a simulator.

Zhou et al. apply a related idea to quadrupedal loco-manipulation using a hierarchical policy that predicts both end-effector trajectories and the tactile signals expected to develop during the task [39]. These predictions are then tracked by a lower-level whole-body controller. Their results show improved performance compared with vision-only methods and simpler visuotactile baselines. This suggests that tactile behaviour can be specified as part of the task command rather than treated only as feedback after contact occurs. In tool-use tasks, such information may help identify handle slip or changes in the grasp even when the tool tip or contact surface cannot be seen clearly.

Each sensing method therefore offers a different compromise. A force/torque sensor gives the clearest measurement of interaction forces and provides a reliable basis for safety checks and validation. Proprioceptive estimation avoids additional hardware, while tactile sensing provides information about local pressure, deformation and slip. For this thesis, the approach will be to implement proprioceptive force estimates to reduce dependence on force sensors and to more easily support a diverse usage of the end effector. Force sensing may be applied in an evaluative capacity to assess the effectiveness of the policies.

2.6 External Tool Use in Robot Manipulation

External tools expand a robot's capability by changing the geometry through which it interacts with the environment. A tool may extend reach, concentrate force, convert motion into a different contact action or access a region that the bare gripper cannot reach. In this thesis, tool augmentation refers to a distinct object grasped or attached to the manipulator and used through a functional tool tip. This differs from direct manipulation by the gripper and from a permanently integrated end effector whose geometry and dynamics never change.

The tool introduces a new transform between the gripper and the functional contact point. Position and orientation commands must therefore be expressed at the tool center point, while forces applied at the tool tip generate additional moments at the wrist and robot base. Tool mass, centre of mass and inertia

also alter the dynamics of the arm, and the grasp may add compliance or slip. A controller that ignores these changes may accurately track the gripper while producing incorrect tool-tip motion or destabilizing the quadruped during contact.

Liu et al. jointly learn a task-conditioned tool designer and a design-conditioned controller, demonstrating that tool geometry can be treated as a policy input rather than only as a fixed mechanical attachment [16]. Their results expose a design-control trade-off: a more capable tool can simplify the required motion, while a limited tool places greater demands on control. Although the tools in this thesis are specified rather than generated, the same principle means that task difficulty cannot be compared fairly without considering tool length, contact shape and mechanical advantage.

Tool use also creates a data-collection problem because the relevant motion is tool-centric rather than robot-centric. Tool-as-Interface transfers human demonstrations to a robot by using shared physical tools, segmented observations and task-space tool actions [3]. This reduces dependence on slow robot teleoperation and explains that demonstrations can remain useful across changes in camera viewpoint and robot configuration when they are represented around the tool. For the proposed system, the same principle supports collecting desired tool-tip trajectories independently of the quadruped and allowing the whole-body controller to resolve embodiment-specific motion.

SimToolReal identifies three distinct difficulties in practical tool manipulation: acquiring thin objects from a surface, reorienting them into functional configurations and maintaining a stable grasp during forceful interaction [12]. A single policy trained on procedurally generated tool-like primitives is conditioned on object-centered goals and transfers across several real tool categories. The important implication is that successful pickup does not imply successful tool use. Acquisition, reorientation, registration and forceful execution should be evaluated separately so that failures can be attributed to the correct part of the system.

ToolTango operates at a higher planning level by predicting which tool should be used and which symbolic action should follow, including cases where a familiar tool is unavailable and an alternative is present [33]. This is relevant to future autonomous selection, but it is intentionally outside the central control problem of this thesis. Specifying the tool in advance isolates whether the quadrupedal controller can acquire and use it with suitable pose, force and compliance.

Furthermore, a permanently mounted or automatically changed tool removes much of the grasp uncertainty. This thesis will implement an assisted tool change from a known position to do away with the acquisition, registration and slip errors a freely stored external tool introduces, and will define the success of tool use on a per task basis.

2.7 Tool-Conditioned and Multi-Tool Policies

A controller trained with only one tool may develop a policy strongly connected to tool geometry without representing it explicitly. This becomes a problem when the robot changes to a tool with a different length, mass or payload. Supporting several tools therefore generally requires either an individual policy for each tool or one shared policy that receives information about the tool currently in use. Specialist policies avoid confusion between different tool dynamics, but the number of policies grows with every new tool and little knowledge is transferred between them. A shared policy is easier to maintain, although free of explicit conditioning it must determine the active tool, behaviour and its functional frame from the available observations.

A discrete tool identifier is the most direct way to provide this information. It allows the policy to distinguish between tools encountered during training, but it says nothing about how those tools are

related. It is also unlikely to support direct generalisation to a tool with an unknown shape. Liu et al. demonstrate a more information-driven approach by conditioning the controller on both the generated tool design and the task objective [16]. In the proposed system, the input could instead describe physical properties such as the tool-tip transform, mass, centre of mass, inertia, overall length, contact dimensions and expected interaction type. Unlike a simple label, these variables indicate the physical reasons that the controller may need to behave differently.

Another possibility is to represent the task relative to the tool itself. Tool-as-Interface encodes demonstrations as task-space actions of the tool, which reduces their dependence on a particular robot embodiment [3]. SimToolReal uses the current and desired tool poses together with a rough geometric description of the graspable area to condition an object-centred policy [12]. Because this representation does not depend on a one-shot identity, it can be applied to real tools that were not present during training. A descriptor based on pose and geometry may therefore offer better generalisation than a discrete tool label, although its success depends on estimating the tool state accurately.

High-level semantic depictions become more relevant when the robot must decide which tool to use or find a substitute for one that is unavailable. ToolTango uses knowledge-based object representations to adapt symbolic plans when a familiar tool is missing [33]. Generalist policies such as Octo and OpenVLA also provide broad visual and task-level representations [6], [13]. These systems, however, do not show that a large visual policy can replace the fast feedback required for balance and force regulation. Within this thesis, they are more plausibly treated as future perception, tool-selection or task-planning components than as the main whole-body controller.

There may also be several valid ways to complete the same tool-use task, with differences in approach direction, grasp choice or tool reorientation. A diffusion policy at the higher level could represent these alternatives, while a deterministic lower-level controller maintains balance and satisfies contact constraints [4]. Any evaluation of this design should still isolate the effect of tool conditioning. The conditioned and unconditioned policies should use the same observation set, task distribution, training data and optimisation budget. Otherwise, an apparent improvement could result from additional model capacity or training exposure rather than from the tool representation itself.

2.8 Contact-Rich Manipulation Tasks and Performance Evaluation

Contact-rich manipulation should be evaluated through repeatable task definitions rather than individual successful demonstrations. Zhi et al. combine repeated empirical trials with continuous force and position errors, showing the value of reporting both task outcome and controller behaviour [38]. This is important because success alone can hide unsafe or poor-quality execution, where excessive force, repeated contact loss or an unstable whole-body posture may have contributed to a success.

Suitable controller metrics consist of tool-tip position error, force root-mean-square error, peak-force overshoot, time within the desired force band and number or duration of contact losses. Whole-body performance should also include falls, non-foot collisions, joint or torque-limit violations, action smoothness and energy use. Task-specific quality measures are also needed. For example, wiping can be assessed by coverage, scraping by removed material or path completion, and pressing by successful mechanism activation without exceeding a damage threshold.

Capability measures can reveal why a policy succeeds or fails. QuadWBG reports reachable workspace and grasp success in addition to individual tasks [35]. For tool-mediated manipulation, corresponding measures include reachable tool-tip workspace, sustainable force direction and magnitude, grasp stability and the range of base poses from which the task can be completed. Kang et al. further

demonstrate that grasp retention and object damage must be separated when force-sensitive objects are involved [11].

Generalisation should be tested using explicit training and evaluation splits. ManiSkill2 uses diverse task families and object geometries, demonstrating the value of holding out object instances or shapes rather than evaluating only the exact models seen during training [7]. LIBERO similarly distinguishes different forms of knowledge transfer [14]. A multi-tool experiment should therefore separate generalisation to a new tool instance, a changed surface or contact property, and a new tool trajectory, because each tests a different capability.

Overall, the literature indicates that contact-rich manipulation should not be evaluated through binary task outcome alone. A suitable evaluation should combine task outcome with continuous measures of force and position tracking, contact stability, whole-body safety and task-specific quality. Generalisation should also be assessed using explicit training and evaluation splits that distinguish changes in tool geometry, contact properties and task trajectory. These principles inform the experimental protocol developed in Section 4.4.

2.9 Simulation-to-Real Transfer

Reinforcement-learning policies for legged robots are typically trained in simulation because physical training is slow, expensive and unsafe. Hwangbo et al. established that learned dynamic motor skills can transfer to real quadrupeds when simulation, actuator modelling and robust training are treated as central parts of the method [10]. Contact-rich tool use increases this difficulty because the policy depends not only on the dynamics of locomotion but also on arm actuation, contact compliance, tool properties and force observability.

Dynamics randomisation addresses modelling error by training over a distribution of physical parameters so that the real robot behaves like another member of the training set [23]. Common parameters include mass, inertia, friction, motor strength, damping, sensor noise and control delay. Tool use adds the tool mass and centre of mass, grasp transform, handle compliance, surface friction and contact geometry. Randomising these quantities can prevent the policy from depending on one ideal simulated tool, but ranges that are unrealistically broad may reduce precision or encourage overly conservative behaviour.

Xie et al. caution that dynamics randomisation is neither automatically necessary nor sufficient for quadrupedal transfer [36]. Their ablations show that action representation, control design and other modelling choices can be as important as randomisation. This supports a sensitivity-driven approach: sim-to-sim tests should identify which actuator, contact and tool parameters materially affect force and task performance before selecting randomisation ranges. System identification should be used where feasible to narrow the nominal model and avoid asking one policy to cover unnecessary uncertainty.

UMI-on-Legs provides direct evidence of the importance of model correctness for a quadrupedal manipulator [8]. The authors found that the manufacturer's arm model was insufficient for dynamic tasks and that remeasuring component mass, centre of mass and inertia was vital to stable simulation and physical deployment. They also found that approximately 20 ms of simulated end-to-end control latency best matched their real system, while mismatch produced high-frequency shaking. The additional latency of the pose estimator and software callbacks further demonstrates that sensing and control rates must be modelled separately.

Safety-related behaviour may also change considerably between simulation and hardware. When simulation-trained UMI-on-Legs checkpoints were deployed directly, the authors observed shaking,

joint overheating and unsafe twisted postures [8]. These problems were reduced by increasing action-rate and torque regularisation and by rewarding balanced body postures. Such terms should therefore not be regarded only as penalties for visually undesirable motion. They directly affect whether the policy can function safely and continuously on the physical platform.

Force estimation poses an additional transfer problem. Zhi et al. report axis-dependent errors caused by mismatch in actuator dynamics as well as contact modelling [38]. A tool can amplify such errors because a small wrench error at the wrist may correspond to a larger tool-tip position or moment error. Simulation evaluation should therefore report force error by axis, workspace region and tool configuration rather than only as a single average.

A number of strategies can reduce visual and geometric transfer demands. SimToolReal procedurally varies tool-like geometry and uses an object-centric goal representation, enabling zero-shot transfer across real tools without task-dependent retraining [12]. Isaac Lab provides the actuator models, multi-frequency sensing and domain-randomisation infrastructure required to implement similar experiments [18]. Nevertheless, simulator version, physics settings, control rate, observation latency and randomisation distributions must be recorded as experimental parameters rather than treated as implementation details.

Considering this, a staged deployment protocol will be implemented. The policy should first be validated for joint and base tracking without environmental contact, followed by low-force bare-end-effector interaction, a lightweight pre-attached tool and finally mechanically assisted tool acquisition (from a pre-defined position) and full task execution. Each stage should use bounded commands, conservative force and torque limits, an accessible emergency stop and predefined abort conditions.

2.10 Summary of Research Gaps

As summarised in Table 2, existing research demonstrates the individual capabilities required for the proposed system, including learned whole-body control [5], [8], [15], [38], force-aware manipulation [24], [38], and tool-conditioned or tool-centric control [3], [12], [16]. However, considerably less evidence exists at their intersection, particularly where force-position whole-body control is applied through interchangeable external tools. The following gaps therefore concern the integration and controlled comparison of these previously demonstrated capabilities.

Table 2: Summary of research papers and their limitations

Work	Quadrupedal whole-body control	Force explicitly controlled	External tool-mediated contact	Explicit tool conditioning	Real robot	Main limitation relative to this thesis
Fu et al. [5]	✓	—	—	—	✓	Whole-body coordination but position-based
Portela et al. [24]	✓	✓	—	—	✓	Position/force modes are separated and contact is at the end effector
Zhi et al. [38]	✓	✓ unified	—	—	✓	Unified force-position control, but not through changing external tools
UMI-on-Legs [8]	✓	—	Limited	—	✓	Good hierarchical and sim-to-real framework, but no explicit force regulation

Schperberg et al. [29]	✓	✓ compliant /	—	—	✓	Hybrid/model-based architecture rather than the proposed unified learned force-position policy
Liu et al. [16]	—	task dependent	✓	✓	— /limited	Demonstrates design-conditioned tool control outside quadrupedal WBC
SimToolReal [12]	—	—	✓	geometric/object-centric	✓	Tool generalisation without quadrupedal balance or explicit force-position WBC
Zhou et al. [39]	✓	tactile-aware	—	—	✓	Contact-aware WBC, but not explicit multi-tool force-position control

The first gap is whether the benefit of unified force-position control is retained when contact is displaced through an acquired external tool. A tool changes the functional frame, grasp transform, payload, inertia and direction through which force is transmitted. Existing legged force-control studies predominantly evaluate direct end-effector contact, while tool-use studies generally focus on fixed-base or dexterous-hand systems rather than whole-body quadrupedal balance.

The second gap concerns the policy representation required for multiple tools. Prior work supports discrete task conditioning, design conditioning, tool-centric actions and object-centred geometric goals [3], [12], [16], [33]. However, there is limited evidence comparing an explicitly tool-conditioned whole-body force policy with an equivalent unconditioned policy. It remains unclear whether a simple identifier is sufficient or whether physical properties and the tool-tip transform are needed for generalisation.

The third gap is force observability during tool-mediated interaction. Proprioceptive estimation [24], [38], direct wrist force/torque sensing [29], reactive tactile force control [11] and tactile-aware quadrupedal policies [39] provide different balances among accuracy, hardware complexity and transfer. Their suitability for regulating force at a displaced tool tip on a small quadruped has not been resolved.

The fourth gap is simulation-to-real transfer under changing tools. Tool mass, centre of mass, inertia, contact geometry, friction, grasp compliance and latency introduce mismatches beyond the standard locomotion transfer. The literature indicates that randomisation should be supported by model identification and sensitivity analysis instead of being applied indiscriminately [8], [36].

This thesis can address these research gaps by investigating

1. force-aware versus position-only control,
2. tool versus end effector,
3. tool-conditioned versus unconditioned control,
4. simulation versus physical deployment

Together, these experiments can determine whether a hierarchical tool policy and unified low-level force-position controller provide a measurable advantage for contact-rich quadrupedal tool use.

3 Research Question and Project Plan

3.1 Research Question

The central research question of this thesis is:

Can a force-aware whole-body reinforcement learning controller enable a quadrupedal mobile manipulator to perform contact-rich manipulation tasks using specified external tools?

In this project, *force-aware* refers to a controller that receives force-related commands and uses force information or estimates alongside position objectives, while *whole-body* refers to coordinated control of the quadruped legs, floating base and D1 arm. Contact-rich manipulation will focus on sustained-contact tasks such as pressing, wiping, scraping and low-torque lever operation, where successful execution depends on both tool-tip motion and interaction force. The external tools will be specified and registered to the controller rather than autonomously discovered or selected. The research question will be addressed through controlled comparisons between force-aware and position-only control, tool and bare-end-effector operation, tool-conditioned and unconditioned policies, and equivalent simulation and physical trials.

3.2 Hypotheses

- H1 - Force-aware control will achieve higher task success and lower contact-force error than a position-only controller on contact-rich tasks.
- H2 - Using a task-appropriate external tool will improve task success or efficiency compared with the bare end-effector.
- H3 - A tool-conditioned policy will generalise across multiple tools more effectively than a non-tool-conditioned policy trained under otherwise equivalent conditions.
- H4 - Domain randomisation and appropriate system modelling will enable transfer of the selected force-aware policy from simulation to the physical robot with an acceptable performance reduction.

3.3 Project Plan

The first step in this project is constructing a simulated environment where training can be undertaken. This will be done with ROS 2 Humble on Ubuntu 22.04 running Isaac Sim 5.1 with Isaac Lab 2.3 [19]–[22]. This version of Isaac Sim and Lab supports ROS 2 Humble integration and through its Python Omniverse API or OmniGraph I can enable the locomotion and arm to accept `cmd_vel` TWIST Teleop and pose messages as an optional alternative to a top commanding policy. Preparation for the Sim is mostly complete and involved defining the URDF model of the robot and setting up its inertial and joints.

The project will follow a baseline-first development strategy where each additional capability will be introduced only after the preceding stage has met predefined validation criteria. The simulation environment and position-only controller will first establish that locomotion, reaching and task execution are reproducible. Force-aware control will then be introduced and evaluated before tool conditioning is added. This staged approach allows performance changes to be attributed to individual design choices and reduces the risk of attempting physical deployment before the controller is stable in simulation. The work stages and deliverables are summarised in table 3 below:

Table 3: Staged project work packages and decision gates.

Work stage	Main activities	Deliverable / decision gate
WP1 Baseline simulation	Build robot/tool models and position-only baseline; validate locomotion and reaching	Stable baseline and reproducible task environment
WP2 Force-aware policy	Implement force commands/estimation, rewards and curriculum	Force-aware policy meeting simulation thresholds
WP3 Tool conditioning	Implement tool descriptors and train multi-tool variants	Conditioned and unconditioned comparison
WP4 Ablation experiments	Run controlled comparisons with fixed seeds and budgets. See appendix C for more detail on these tests.	Complete simulation results and analysis
WP5 Sim-to-real transfer	System identification, randomisation tuning, safety tests and staged deployment	Physical robot demonstrations and transfer results

Four initial contact-rich tasks have been selected to evaluate the proposed controller: pressing, wiping, scraping and low-torque lever operation. These tasks provide different combinations of position tracking, sustained normal force and tangential or rotational motion. Detailed task definitions, including success and failure criteria, can be seen in Table 4:

Table 4: Initial contact-rich tool-use task definitions.

Task	Tool condition	Commanded interaction	Success and primary measures	Failure conditions	Domain
Pressing	Rounded probe or bare end effector	Approach in position control, then regulate force normal to a spring-loaded button	Button reaches full travel and remains engaged for 1 s; report success, force RMSE, peak force and time	No activation, excessive peak force, slip, collision or fall	Simulation and real
Surface wiping	Microfibre pad	Track a planar path tangentially while maintaining normal force	At least 90% path completion and contact maintained for at least 80% of the motion; report force RMSE and coverage	Contact loss, path departure, excessive force or unstable posture	Simulation and selected real trials
Resistive scraping	Plastic scraper or spatula	Track a line while resisting tangential load and maintaining tool orientation	At least 90% path completion with no more than one brief contact loss; report completion time and force error	Tool lift-off, rotation out of tolerance, stall, collision or fall	Simulation and selected real trials
Lever rotation	Lightweight wrench or lever tool	Maintain engagement while producing a commanded rotation about the fixture axis	Fixture reaches the commanded angle and the tool remains registered; report angle error, completion time and peak force	Tool disengagement, excessive wrist load, fixture impact or loss of balance	Simulation first; real if safe

The proposed system will use a hierarchical architecture, with a high-level task interface providing the desired tool-tip trajectory, interaction force and information about the attached tool to a lower-level whole-body controller. The lower-level controller will be trained using reinforcement learning to coordinate the Go2 legs, floating base and D1 arm while maintaining balance and following the required position and force commands. This follows the literature reviewed in Sections 2.2–2.5, where hierarchical control is useful for separating slower task-level decisions from the faster whole-body control required for locomotion, contact and force regulation. The proposed architecture is shown in Figure 1 below.

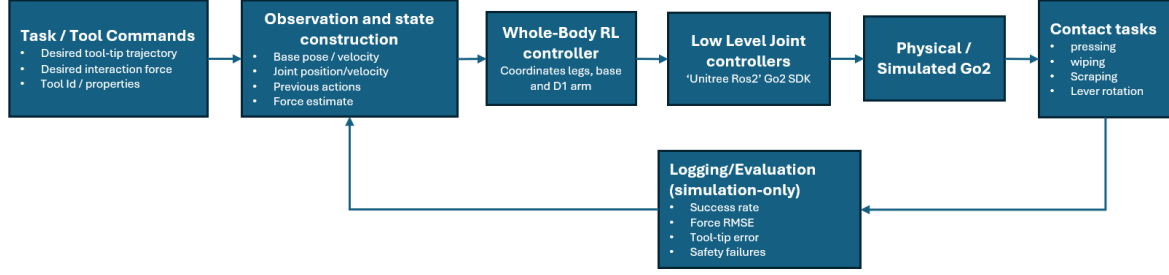


Figure 1: Proposed hierarchical control architecture for tool-augmented force-position control.

The policy observation will include the robot's proprioceptive state, task commands and previous actions, with force estimates and tool information added depending on the policy being tested. Tool-related commands will be defined at the functional tool tip rather than the gripper so that changes in tool length and contact location are represented in the control problem. The policy will output whole-body joint targets that are tracked by the robot's lower-level joint controllers. This same general architecture will also be used for the position-only and unconditioned baselines, with the relevant force or tool inputs removed so that the effect of each addition can be compared without changing the rest of the system. Development will initially take place in Isaac Sim before the selected controller is transferred to the physical Go2-D1 platform. The simulation and physical implementations will retain the same command and observation structure where practical, while system identification, actuator modelling and targeted randomisation will be used to account for differences between the two systems. Detailed scheduling, task overlap, milestones and contingency periods are provided in the Gantt chart in Appendix A.

3.4 Required Training and Upskilling

Previous experience with the Go2 and D1 arm, and extensive work with ROS 2, Isaac Sim, and Python have provided me with the confidence and baseline to approach this thesis. However my exposure to RL, especially with manipulators is lacking beyond the theory and research conducted thus far. Upskilling will primarily involve becoming confident in writing, assessing and evaluating RL policies so that in particular I can synthesise the lesser documented approaches seen in the papers I have analysed. Practical experience with sim-to-real and collecting demonstrations for the high-level autonomy is another area requiring moderate upskilling, as my experience with visual trajectory policies is minimal. Training has started by attending the RimA summer robotics school at FZI Karlsruhe Germany, which focuses on RL policies and autonomous household interactions. The learnings so far have been very valuable and tailored also to tool usage, and the advice from professors and industry contacts at Flexion robotics helped inform some architecture choices. The robots I am using at this school can be seen in figure 2:

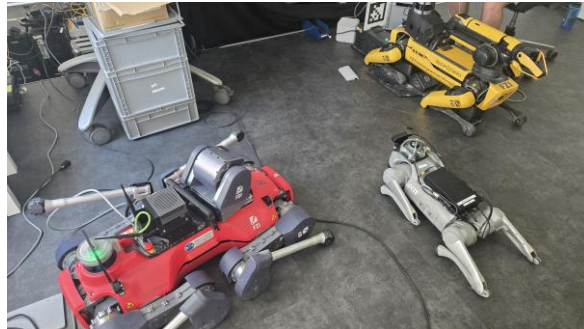


Figure 2: ANYmal, Go2, and Spot at FZI Karlsruhe.

Furthermore, training is continuing through online tutorials on setting up and customising RL policies in Isaac Lab. In general too, investigation into the concepts described in the papers from section 2 is continuing to further assist my understanding.

4 Project-Dependent Preparations

4.1 Existing Platform and Software Preparation

The physical platform, Unitree Go2 EDU and D1 arm are available in the Ainsworth J17 UNSW robotics laboratory, and also as personal property at home (acquired through my startup). The arm mounting hardware and manufacturer model files have been obtained directly from Unitree, and the combined robot has been represented in Isaac Sim with the arm positioned above the Go2 expansion dock. Development is being undertaken on Ubuntu 22.04 using ROS 2 Humble, CycloneDDS and the Unitree D1 software interface so that the same message structure can be retained between simulation and hardware where practical. The current simulation preparation includes the combined articulation, joint-state publication, arm command interface and L1 LiDAR output. An example fixture for the lever task, a lightweight wrench tool and associated combiner box (an example of a solar farm inverter box) has been acquired for testing.

4.2 Preliminary Results

As of this report, progress has been made to import the acquired URDF models into Isaac Sim 5.1 and verify that the inertials and masses do align between the Go2 and D1 arm. The arm has been mounted in the appropriate location corresponding to its mounting rails above the real Go2 EDU expansion dock, and has been tested with an existing locomotion controller. The results were as expected for an RL policy designed without payloads in mind - the robot struggled more during actions and fell over particularly in response to sudden sideways movement. This proves the need for a locomotion policy to be contextually aware of a mounted mobile manipulator, whether by unified whole body control or by explicit parameterised input. The assembled simulation model and arm mounting arrangement are shown in Figure 3.

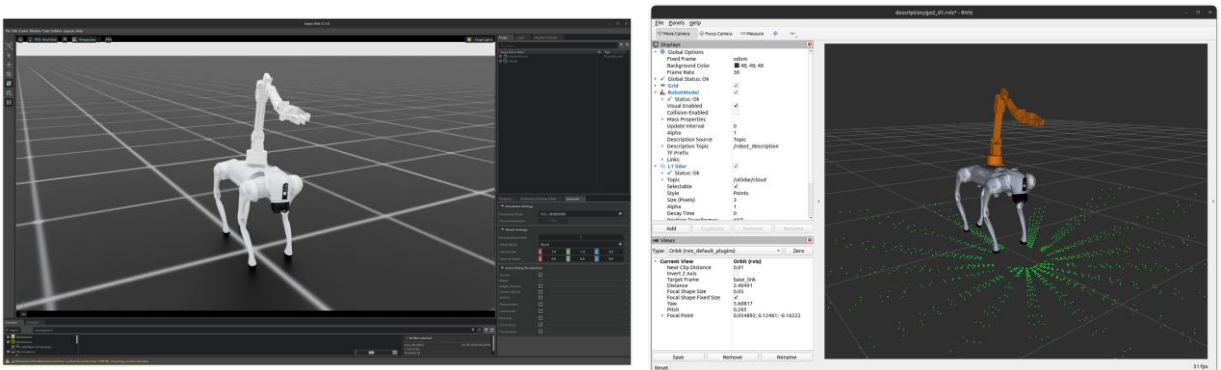


Figure 3: Go2-D1 model in Isaac Sim 5.1 (left) and simulated state with L1 LiDAR visualised in RViz2 (right).

The simulation interfaces with ROS 2 and publishes joint positions, the built-in Unitree L1 LiDAR, and importantly is also bundled with a CycloneDDS D1 SDK from Unitree that acts as the API for the arm. This is a first step towards ensuring the sim to real gap is closed as much as possible, so that deployment faces less hurdles particularly interface-wise. This can be seen in the RViz2 window in Figure 2, which displays the LiDAR, joints and meshes for the arm and for the Unitree Go2, synced with Isaac Sim 5.1.

4.3 Component List

Table 5: Tool and component list.

Quantity	Component / tool	Purpose	Available?	Cost / source
1	Unitree Go2 EDU	Quadrupedal mobile base and whole-body locomotion platform	Yes	Existing UNSW laboratory equipment
1	Unitree D1 arm	Arm manipulation and tool positioning	Yes	Existing UNSW laboratory equipment
1	Intel Realsense D435i	Visual sensor for high level arm policy	Yes	Existing UNSW laboratory equipment
1	Tool mounting interface	Repeatable attachment, retention and tool-frame registration	No	Custom fabricated from FDM print PLA
1	Force/torque measurement device or instrumented fixture	Ground-truth calibration and evaluation only	Yes	Personal Inventory
1	Rounded pressing probe, Microfibre wiping pad, Plastic scraper,	Point-contact force regulation and button-pressing tasks	No	Custom fabricated from TPU95a FDM filament using a 3d Printer
1	Lightweight wrench and combiner box	Apply controlled torque to a rotary handle, valve or lever fixture	Yes	Personal Inventory

4.4 Risk Assessment

Below is a risk assessment specific to RL testing with tool-based tasks and a sim-to-real pipeline. Risks associated with general laboratory safety can be found on the UNSW Salus portal [34].

Table 6: Project-specific risk assessment.

Risk	Likelihood (1-5)	Controls
Robot collision with a person	2	Cordon area and ensure a 1-2m distance to the robot at minimum; Apply speed/torque limits and have an accessible emergency stop;
Excessive tool/contact force damages robot, tool or workpiece	2	Software force limits; compliant fixtures; ground-truth monitoring; abort conditions
Robot collides with environment	3	Test in middle of Ainsworth J17 robot lab soccer field with plenty of open space; low-gain/low-speed trials; bounded actions; safety supervisor; incremental deployment
Insufficient training time or compute	3	Baseline-first plan; checkpointing; reduced task matrix; pre-defined scope decision gates
Robot falls during whole-body manipulation	3	Screen policies in simulation; use a fall-safe area and support rig where appropriate; begin from conservative postures and low speeds; stop after repeated instability
Tool detaches or is incorrectly registered	2	Use positive mechanical retention; inspect the mount before each trial; verify tool ID and transform; use lightweight tools and bounded payloads
Sim-to-real policy behaves unexpectedly	3	Validate commands in shadow or disabled-actuation mode; use low-gain and low-speed trials; bound actions; maintain a safety observer and immediate emergency-stop access

5 Conclusion

This report examines the control problem that arises when a quadrupedal mobile manipulator is required to carry out sustained-contact tasks using an external tool. As shown in the literature, learned whole-body control is able to coordinate locomotion, posture and arm movement, and force-aware policies can extend position tracking to include direct force control and compliant interaction. Research into tool use also highlights the importance of the functional tool frame, as well as the geometry and control that is conditioned by the design of the tool. The outstanding issue remains the controlled integration of these approaches on a quadrupedal platform, especially in regard to whether force awareness and explicit knowledge of the tool still prove beneficial when the point of contact is shifted through the mounted tool.

To deal with this gap the project will make use of a Unitree Go2 EDU and a D1 arm. The central research question will be examined by means of four comparisons: force-aware control compared with position-only control, task performance with and without a tool, tool-conditioned policies as against unconditioned policies, and simulation compared with physical implementation. Performance will be evaluated using the repeatable task criteria along with force error, peak force, tool-tip position error, contact loss, completion time and any safety-related failures. The tasks are planned stress controlled pressing, wiping, scraping and the use of levers at low torque rather than high-impact or unrestricted tool use.

So far the preparation has involved implementing the combined Go2-D1 model in Isaac Sim 5.1, adding the ROS 2 and D1 arm interfaces, providing LiDAR and joint-state visualisation, and carrying out preliminary locomotion tests with the arm attached. The fact that a locomotion policy trained without taking the arm payload into account shows degradation reveals that it is necessary to consider the full robot configuration when developing the controller. The next stage will be to achieve a stable position-only baseline and to establish a reproducible task environment, then move on to force-aware policy training and a controlled tool-conditioning comparison; physical deployment will not start until the selected policy has successfully passed the various stages of simulation and the low-force safety checks.

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Appendix A - Gantt Chart

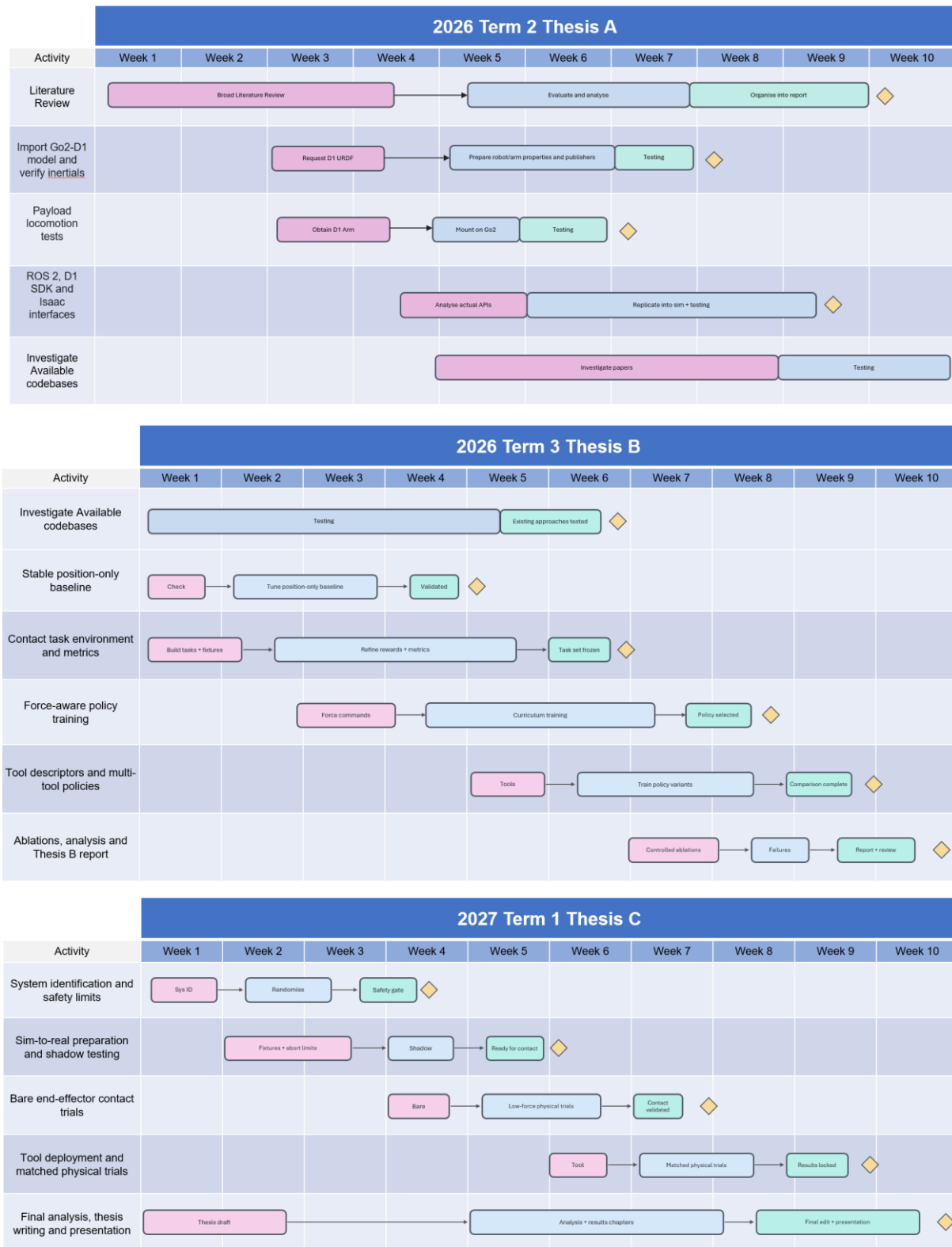


Figure 3: Provisional schedule for the remaining Thesis A, Thesis B and Thesis C work.

Appendix B - Draft Final Thesis Outline

1. Introduction

- Motivation and Background
- Research Objectives
- Research Question and Hypotheses
- Thesis Outline

2. Literature Review

- Quadrupedal Mobile Manipulation and Whole-Body Control
- Position, Force and Impedance Control
- Reinforcement Learning for Contact-Rich Manipulation
- Force Sensing and Estimation
- External Tool Use and Tool-Conditioned Policies
- Simulation-to-Real Transfer
- Summary of Research Gaps

3. Methodology

- Development of the Simulation and Robot Model
 - Unitree Go2 EDU and D1 Arm Model
 - External Tool and Task Models
 - ROS 2 and Isaac Lab Interfaces
- Development of the Whole-Body Controller
 - Position-Only Baseline
 - Force-Aware Controller
 - Tool-Conditioned Controller
- Development of Contact-Rich Tasks
 - Pressing
 - Wiping
 - Scraping
 - Lever or Wrench Operation
- Simulation-to-Real Transfer
- Experimental and Evaluation Procedure

4. Results and Discussion

- Force-Aware versus Position-Only Control
- Tool versus Bare End Effector
- Tool-Conditioned versus Unconditioned Control
- Simulation versus Physical Robot Performance
- Failure Analysis and Limitations
- Future Work

5. Conclusion

Appendix C - Experimental Matrix

The policy matrix fixes the main ablations before large-scale training. Policies P2-P4 will use the pressing probe, wiping pad, scraper and lever tool unless the task set is reduced at a documented scope gate.

Table C.1: Experimental policy matrix.

Policy ID	Force-aware?	Tool present?	Tool-conditioned?	Training tools	Evaluation domain	Purpose
P0	No	No	No	None	Simulation + real where safe	Position-only baseline
P1	Yes	No	No	None	Simulation + real	Effect of force awareness without tool
P2	No	Yes	No	Probe, pad, scraper and lever	Simulation + real	Effect of tool under position-only control
P3	Yes	Yes	No	Probe, pad, scraper and lever	Simulation + real	Force-aware tool policy without explicit conditioning
P4	Yes	Yes	Yes	Probe, pad, scraper and lever	Simulation + real	Full proposed method